

Trading Engine - active checklist

Updated **18 September 2026**, following the source review at `6c2f2f6`. This is the single active task list. [Plan v4](#) owns long-range scope; [the guide](#) explains the work and effort estimates.

[TODO.pdf](#) is an export of this checklist. The originally supplied [new-todo-list.pdf](#) is historical input, not the current plan. Documentation edits do not complete the code tasks below. No deadlines are agreed.

Start here

Next learning task: D1, agree the accounting examples; then D2, implement Portfolio. The original foundation repair batch closed on 16 September. ADR 0014 was accepted on 15 September; do not reopen all eight decisions. The review found a narrower D5 accounting gap and additional hardening work, listed separately below.

Recommended impact/effort order: ~~C1 sanitizer enforcement~~ (done 18 September), **D1 accounting rules next**, then C2 capture admission, C3 tape preconditions, then the complete D2-D7 engine path. In-memory Portfolio work can proceed while capture hardening remains open. C2 gates trusted capture-to-engine results; C3 gates tape use.

Fuaad chooses domain rules and writes the first learning-critical implementation. The assistant writes all tests against agreed examples and explains their purpose. Tests must not silently select open accounting or simulation policy.

A. Preserve data and reproducibility

✓ **Capture recovery/backup:** user reports the backup task complete. Earlier recovery checks found no remaining `dataless` files. An independent byte comparison of the backup was not performed; do not turn that distinction into a repeated blocker.

✓ **Broken build directories:** repaired on 16 September. The recorded cause was iCloud conflict duplicates. Do not repeat the old deletion instructions.

✓ **Build outside the synced Desktop tree.** Done 21 September 2026. The canonical build is now `~/build/TradingEngineProject`, outside iCloud sync:

```
bash cmake -S . -B ~/build/TradingEngineProject -DCMAKE_BUILD_TYPE=RelWithDebInfo cmake --
build ~/build/TradingEngineProject -j ctest --test-dir ~/build/TradingEngineProject --
output-on-failure
```

Measured, not assumed: reconfiguring the durable build takes **2.8 seconds**; the same command against the in-tree `build/` timed out at **5 minutes**. 327/327 pass there.

This also fixes editor tooling. `compile_commands.json` is copied from whichever directory you build in, so building anywhere transient points clangd at paths that later vanish and every `#include` goes red. Build in the durable directory and the file stays valid. The in-tree `build/` still exists and still resyncs;

prefer deleting it so it cannot silently become the source of `compile_commands.json` again. - [x]

Previously pending work: present in history at the review baseline; the checkout was clean before this documentation update. New changes remain uncommitted until requested.

B. Design status

✓ **ADR 0014 D1-D8 accepted, 15 September.** Stage 5 decision inputs are book trust and market shape; operational state belongs to Stage 9.

✓ **ADR 0015 accepted, 16 September:** fatal allocation failure, unsafe object, explicitly interim. This does not mean rollback has been implemented.

- **Complete D5's signed-position accounting policy** through D1 below. The 18 September ADR correction identifies the missing policy; it does not choose it.
- **Choose Strategy dispatch when implementing the seam.** Compare runtime and static alternatives in ADR 0009. Do not assume templates or `std::function` are inherently faster.

C. Additional hardening from the 18 September review

C1. Make sanitizer findings fail CI

✓ **Done 18 September 2026.** `cmake/sanitizers.cmake` now adds `-fno-sanitize-recover=undefined` whenever `TE_SANITIZE` includes `undefined`.

The gap was measured, not assumed. A signed-overflow probe built under the previous configuration printed `runtime error: signed integer overflow` and **exited 0**; with the flag it terminates nonzero before reaching its own `printf`. So UBSan was detecting real UB and letting the run stay green — detection without enforcement.

A compile/link flag rather than `UBSAN_OPTIONS=halt_on_error`: the environment variable binds only where it is exported, so local sanitized builds would have kept recovering silently.

`tests/ub_probe.cpp` holds the deliberate UB. It is guarded by `TE_BUILD_UB_PROBE` (OFF by default), is not registered with CTest, and is never linked into `te_core` or any app, so no intentional UB enters the ordinary suite. A CI step builds it separately per compiler and fails if it exits 0, asserting **nonzero** rather than a specific status, since SIGABRT surfaces differently across platforms.

Scope limit worth remembering: this covers UBSan's checks, which include *signed* overflow. Unsigned wraparound is well-defined in C++, is not UB, and is **not** caught — the explicit boundary tests in `trade_reconciler`, `capture_coordinator` and `price_level` remain the only guard there.

It found a real bug on its first CI run. `Sha256.EmptyInput` failed under both gcc and clang at `sha256.cpp:96:30`: an empty span's `data()` may be `nullptr`, and `memcpy` declares `src` nonnull, so passing it was undefined even with a zero length. Harmless on every real libc, but the compiler may infer from the nonnull attribute that the pointer cannot be null and drop a later check on that basis. That UB

predates C1 and UBSan had been reporting it on every sanitized run since the SHA-256 work landed — it just scrolled past in a green log. Fixed in `a1e1f9c` ; digests unchanged, all eight known vectors still match.

Note this was a *nonnull-attribute* violation, not signed overflow: local runs were clean because the probe only exercised the signed-overflow check. Do not read "the local suite passed" as "the codebase has no UB."

Verified: 327/327 under UBSan with enforcement active (before and after the fix), 327/327 Release with real `gcc-15` (unaffected, the guard skips non-sanitized builds), and CI green on `build (gcc)` , `build (clang)` and `release` at `a1e1f9c` . The ASan+UBSan suite could not be run locally — `gtest_discover_tests` fails against ASan-instrumented binaries on this macOS host, a pre-existing limitation; CI covers that config.

C2. Align Python and C++ capture admission

- **Significant; 1-3 days.** Byte/hash/count checks exist. C++ still does not enforce `status` or `chain_valid` . Agree on continuity, ordinals, seed coverage and control boundaries across `manifest_reader` , `segment_loader` , `capture_validator` and the Python validator. **Done when:** shared valid/invalid fixtures receive consistent decisions; interrupted, failed and gapped intervals are rejected or explicitly quarantined under a documented policy. They must never silently become a trusted continuous segment.

C3. Validate tape inputs before opening output

- **Significant; 2-4 hours.** Ordinary replay checks timestamp ordering; the writer does not. Check both streams, seed/cutoff order and header/capture/supplied-seed agreement under the API contract. **Done when:** timestamps [30, 10] with cutoff 20 and inconsistent metadata are rejected before creating output. Do not silently sort a broken stream to conceal a continuity failure.

C4. Preserve useful errors

- **Minor now; 4-8 hours.** Keep detailed loader/validator/replay causes plus segment/location through `captureCoordinator` ; format them at the application boundary. **Done when:** a corrupt fixture reports what failed and where without debugging into the loader.

C5. Refresh the supported toolchain contract

- **Minor; 2-4 hours plus compatibility work.** Move local/CI Python from EOL 3.9 to a supported interpreter and run all Python tests. Keep deliberate archive hashes and dependency pins. **Done when:** the documented local and CI environments agree and pass. Add a local websocket integration test before treating mocks as proof of library behavior.

C6. Enforce allocation policy before recovery or bindings

- **Integration gate; hours to 3 days.** Read ADR 0015 before adding a catching engine wrapper, Python binding or recovery loop. Choose an enforced fatal boundary or exception-safe rollback with explicit process behavior. **Done when:** injected failures prove the chosen behavior; no retained book can be reused corrupt. Quantity-overflow tests do not cover allocation failure.

C7. Extend financial guards alongside accounting

- ✓ **Done 19 September 2026.** `src/engine/portfolio.cpp` and `src/engine/risk.cpp` are now in the guard's exact-financial set, alongside the headers that were already there. Declaring a type in an `int64` header never stopped the arithmetic in the matching `.cpp` from passing through a `double`, which is precisely where a rounding error would enter.

Listed **before** those files exist, deliberately. The set is matched against files actually walked, so an entry for a missing file is inert until the file appears — which means the guard is in place the day Portfolio is written rather than retrofitted after. Also renamed `isExactFinancialHeader` to `isExactFinancialPath`, since it no longer covers headers only.

Verified both directions: a `double` planted in `src/engine/portfolio.cpp` produces violations and exit 1, while a `double` in `src/research/statistics.cpp` is still allowed, so probabilities and statistics keep floating point. Guard passes on the real tree; 13/13 Python tests pass.

D. Build the Stage 5 engine

D1. Agree the money rules using examples

- ✓ **Done 19 September 2026. Twelve rules selected and recorded in ADR 0014 D5.**

Worked by hand so far, each row checked against equity and against realized-plus-unrealized: open long, partial close, final close, open short, partial cover, final cover, and a long-to-short reversal selling 5 while holding 2.

Selected: signed `int64` money at a per-instrument scale finer than the quote currency's minor unit; total basis plus quantity with the average derived, never stored; net fee convention; fees always moving basis against the trader; classification by signed position; proportional reversal fee split; position zero if and only if basis zero; realized produced only by closing.

Also selected: money at **8 decimal places** below one quote-currency unit, chosen from the smallest representable amount rather than the ceiling — one satoshi at \$100,000 is `$0.001` and rounds to zero in whole cents. Duplicates keyed on the **venue's execution ID**, ignored with no state change and retained for the whole run. Overflow **refuses the fill** with a named error, using a 128-bit intermediate for `price x quantity` and a candidate-then-commit update so a half-changed account is unrepresentable. Partial closes **subtract what left** rather than rebuilding from a rounded average — rebuilding multiplies the rounding error by the remaining quantity and invents money, while subtracting conserves the total by construction and hands back the position-zero-implies-basis-zero invariant for free.

Short-to-long by buying is deliberately not hand-worked — it is the structural mirror of the reversal already done, and belongs in a test rather than a paper row.

Only the fee *schedule* (flat, basis-point, maker/taker) is left, and it is a D2 interface question, not an accounting rule. These are accepted intent; nothing is test-verified.

D2. Implement Portfolio and its fill journal

- The assistant writes tests from D1; Fuaad writes the first implementation. Use distinct signed exact types and checked wider multiplication before rescaling. Calculate a candidate update before committing state; record execution and fee identities. **Done when:** long/flat/short/reversal cases pass; failed and duplicate fills cannot half-change state; journal replay rebuilds the account exactly. Complete C7. Fill accounting does not obtain a mark price or wall clock.

D3. Define intentions, reasons and the strategy boundary

- Add only required value types and read-only, lifetime-bounded book access. Strategy observation consumes successfully processed inputs; decisions return intentions without execution authority. Do not introduce Stage 9's `OperationalState`. **Done when:** trust/shape blocks have distinct tested reasons and `NoopStrategy` uses the real interface without producing an order. Settle ADR 0009 when the seam needs it.

D4. Build the simulated venue and real admission checks

- Support submit, accept, reject, partial/full fill, cancel request and acknowledgement. Count accepted outstanding orders in worst-case exposure; do not assume opposing orders fill together. Implement quantity, notional and resulting absolute-position limits. **Done when:** transitions are independently testable, rejection cannot mutate venue/portfolio state, and a fill can correctly race a pending cancellation.

D5. Implement scheduling and information availability

- Keep one thread and inject feed/clock/strategy/venue dependencies. Preserve metadata for zero, fixed (80 ms) and recorded availability policies. Specify ties and how delayed/out-of-order observations preserve a valid strategy view. **Done when:** timelines prove apply-before-observe, no future information, market-before-own-arrival ties, submission-sequence ties and arrival-time fill eligibility. Stored receipt time is deterministic input, not a calibrated network-delay measurement.

D6. Pass the complete engine proof

- Wire feed, book, health, strategy, gate, admission, venue and portfolio. Close C2 before trusted capture runs and C6 before exception recovery. **Done when:** no-op conservation, a hand-calculated scripted fill, a real unchanged-state risk rejection and ten repeated runs pass. Save partial-fill, outstanding-exposure, duplicate-fill and cancel-race cases. Compare intermediate event/order/account state, not only final totals.

D7. Ship the replay executable and run record

- Connect `apps/replay_main.cpp` to the build and actual engine. **Done when:** one command runs a mandatory fixture without private data and reports counts, trust, shape, blocks/rejections, orders, accounting and fingerprints. Record input hashes, commit/build and dirty state, scales, policies, configuration, exclusions and result hashes. Detailed traces are optional. Refresh this checklist and handoff with actual build/test evidence.

E. After Stage 5: learning and research gates

Planned, not implemented; detailed requirements live in plan v4. Longer-range context, including the PhD bridge, is proposed in [research-extension-plan.md](#) — not accepted, and not a prerequisite for anything here.

- [] **E1. Experiment runner — do this first, immediately after Stage 5.** Pulled forward from Stage 10, because every study below is only worth as much as its reproducibility, and results produced before the harness exists have to be redone afterwards. One command reproduces a result from recorded data, parameters and code version. Plan v4 §4's boundary is unchanged — Python configures, reports and analyses; C++ owns the book, fills, risk and ledger. This applies that rule earlier, it does not rewrite it. Every run records dataset identity and exclusions, parameters and seeds, commit/build identity and dirty state, and results. Start with Python invoking the executable and reading structured output; add bindings only when repeated-call or transfer cost justifies the added lifetime complexity. **Done when:** one command regenerates a comparison report from a mandatory fixture with no private data, and a second machine reproduces the same numbers from the same recorded inputs. Effort: 2-4 days.
- [] **E2. Reproducible failure and recovery experiments.** Inject missing, duplicated, delayed and stale inputs, an overloaded consumer, an interrupted run and an invalid model output — each from a recorded scenario rather than a live accident. Market feeds and process sensors produce the same failure shapes, so this transfers; the recovery rules do not. **Done when:** a recorded failure replays to the same defined response every time, with diagnostics that identify what failed and where, and a recovered state that is consistent rather than merely non-crashing. Effort: 2-4 days.
- [] **L3 evidence before Stage 6/8 claims:** compare order IDs/quantities at intermediate checkpoints, and priority only where known. Keep the L2 digest for depth checks. Effort: 1-3 days.
- **Tape equivalence before replacing raw replay:** resolve pre-seed classifier warm-up, bind source lineage and preserve or explicitly reject unavailable timing policies. Prove raw/tape equivalence for each supported policy. Effort: 2-4 days; format work may add time.
- **Stage 6:** observed-order labels, explicit cancellation/ambiguity/censoring, availability-safe features and transparent baselines.
- **Stage 7:** chronological sessions, label-interval leakage protection, frozen evaluation, calibration, block uncertainty and sensitivity to assumptions. This is the trading uncertainty study: does an observed order fill within a defined horizon? It must beat a stated naive baseline, and its claimed uncertainty must

be checked against observed error rather than asserted. Split by session — splitting neighbouring timestamps leaks the answer and produces a reassuring number that means nothing.

- **Stage 8:** equivalent optimized book, controlled profiles and allocation/cache measurements. Compare bounded SPSC with a mutex queue after the single-thread baseline. Add relevant fuzzing, fault injection and concurrency checks. **Record the optimizations that did not help, and why.** A benchmark report containing only wins is a filtered result, not evidence. Measure tail latency, not just throughput — an average hides exactly the behaviour that matters in execution. Every optimization needs correctness equivalence against the reference book before its number counts.
- **Stage 9:** read-only live feed, recovery, paper lifecycle, operational risk, bounded telemetry and shutdown/failure runbooks. No real-money trading.
- **Stage 10:** thin dashboard, ownership-safe C++/Python boundary, reproducible correctness/performance/research reports and a five-minute demonstration.

Advanced techniques may be contained learning experiments. Promotion to the main engine requires correctness equivalence and measured value. A measured non-improvement still teaches something.

F. Completed foundation evidence

Historical observations, not new test results from this documentation update:

- ✓ Exact types/parsing, clock seam, byte codecs and portable v3 segment I/O.
- ✓ Move-only reference book, ordinary-error atomicity and trust/shape separation.
- ✓ Joined capture, merge/reconciliation, mandatory synthetic fixture and input accounting.
- ✓ Reported joined checkpoint agreement: 0 of 4,533 price levels differed. This does not prove every order's identity or priority.
- ✓ Coordinator byte/hash/count admission, overflow guards and optional-corpus skip repair.
- ✓ All four Python test files in CI, hashed C++ archives, GCC/Clang Debug sanitizer builds, a Release job and explicit debug-only structural validation.
- ✓ ADRs 0014/0015 and prior build-directory repair. Historical logs report 325-case toolchain runs and a later 327-case local run on 16 September; cite configuration/date.

[The guide](#) explains the work and conditional timeframe. [The handoff](#) records current source limitations.